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Summary of Statistical Tests for Random and Pseudorandom Number Generators

Introduction

Random Number Generators (RNGs) and Pseudorandom Number Generators (PRNGs) are fundamental in cryptography. Their output must mimic true randomness—unbiased, unpredictable, and pattern-free. To verify this, the NIST SP 800-22 suite provides 15 statistical tests that identify deviations such as biases, patterns, or structural weaknesses.

Overview of the Test Suite

The test suite evaluates different properties of randomness:

- Frequency and uniformity (Monobit, Block Frequency)
- Pattern behavior (Runs, Longest Run, Template Matching)
- Structural and linear complexity (Rank, Linear Complexity, Serial, Approximate Entropy)
- Spectral and random-walk properties (DFT, Maurer, Cumulative Sums, Excursions)

Summaries of the 15 Tests

1. Frequency (Monobit) Test

Checks if zeros and ones occur about equally in the whole sequence.

2. Block Frequency Test

Divides the sequence into blocks and tests whether each block has a near-50% proportion of ones.

3. Runs Test

Counts sequences of identical bits to detect if the sequence oscillates too quickly or too slowly.

4. Longest Run of Ones Test

Checks whether the longest sequence of consecutive ones in each block matches expectations for randomness.

5. Binary Matrix Rank Test

Forms binary matrices from the sequence and examines their ranks to detect linear dependencies.

6. Discrete Fourier Transform (Spectral) Test

Detects hidden periodicity or repetitive patterns in the frequency domain.

7. Non-overlapping Template Matching Test

Counts how often a predefined pattern appears without overlap, revealing abnormal repetitions.

8. Overlapping Template Matching Test

Similar to above but allows overlapping occurrences, useful for spotting dense, repetitive structures.

9. Maurer's Universal Statistical Test

Measures compressibility; highly compressible sequences are not random.

10. Linear Complexity Test

Determines the length of the shortest LFSR capable of generating the sequence; lower complexity implies predictability.

11. Serial Test

Checks distribution of overlapping m-bit patterns to detect bias or dependence.

12. Approximate Entropy Test

Compares frequencies of m-bit and (m+1)-bit patterns to evaluate unpredictability.

13. Cumulative Sums (Cusum) Test

Treats the sequence like a random walk and looks for significant deviations from expected drift.

14. Random Excursions Test

Analyzes the number of visits to specific states during random-walk cycles to detect anomalies.

15. Random Excursions Variant Test

An expanded version analyzing state visit counts across a wider range of states.

Conclusion

These 15 NIST tests collectively assess a broad spectrum of randomness characteristics. A PRNG that passes them with statistically acceptable P-values can be considered reliable for cryptographic applications, though no test suite guarantees absolute randomness. They serve as a rigorous benchmark to ensure that generated sequences do not reveal exploitable patterns.